

Demonstrating Diesmann 1999

exp087 · 20 August 2026 · Draft

Abstract

A synfire chain is a sequence of neuron pools connected in one direction. A brief packet of nearly synchronous spikes enters the first pool, which may generate a new packet in the next pool. Diesmann et al. (1999) showed that weak or diffuse packets disappear, while sufficiently strong packets can converge toward a stable size and temporal width [1]. Here, each network runs for 100 ms against balanced background activity averaging 10.07 Hz per neuron. A weak diffuse input produces no coherent volley. A broad packet grows from 48 to 95 neurons and sharpens, while an oversized narrow packet settles near the same state. This qualitatively demonstrates stable packet propagation within substantial ongoing activity.

Prior work

Diesmann et al. identified two regimes separated by a boundary in packet state space. Below the boundary, activity dies. Above it, different input packets approach the same stable packet [1]. Our neuron model, background drive, and finite chain differ from the original model, so this is a qualitative demonstration rather than an exact reproduction.

Methods

Methods 1–4 are complete. Every simulation uses one network seed and one fixed background realization.

1. **Define the chain and the pulse packet.** Build 6 feedforward pools of 100 conductance-based leaky integrate-and-fire neurons. Each neuron receives exactly 40 excitatory connections from the preceding pool, with total incoming strength 0.12 μS and delay 1 ms. Independent excitatory and inhibitory background inputs produce ongoing irregular spikes. Adjust inhibitory background strength by pool so asynchronous feedforward activity does not raise the mean rate downstream. The first pool receives an artificial pulse packet through the same feedforward connection rule.

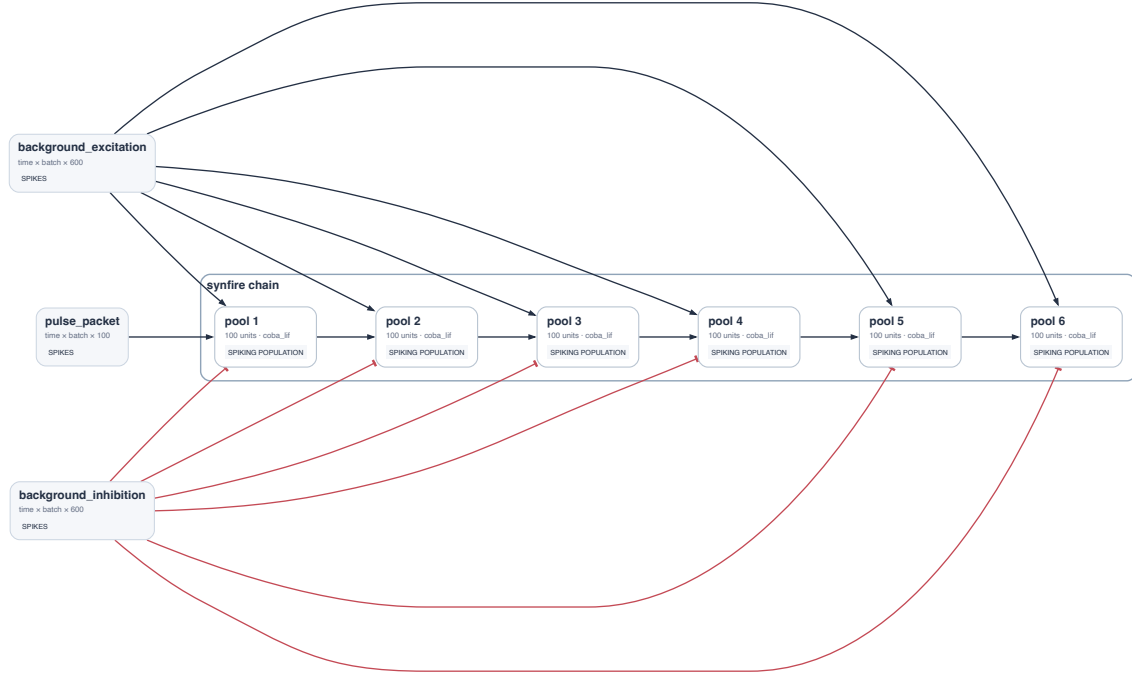


Figure 1: SNNLANG circuit export. A pulse packet enters Pool 1 and activity can travel through six feedforward pools. Independent excitatory and inhibitory background inputs reach every pool.

Describe each packet by its size α , the number of input spikes, and its temporal spread σ , the standard deviation of their spike times. A narrow packet concentrates excitation and is more likely to make the target pool fire. A broad packet distributes the same input over time and may fail to cross threshold.

A pulse packet has a size and a temporal width

Design schematic: α counts spikes; σ measures their spread around the packet centre

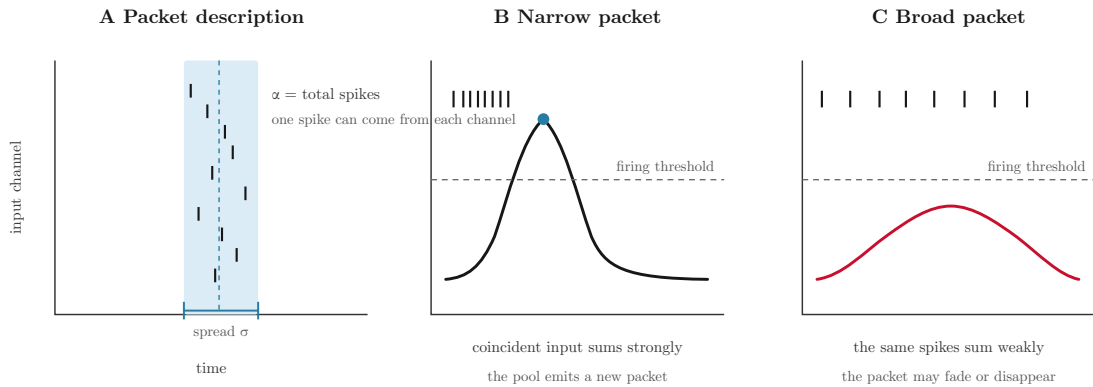


Figure 2: **Pulse packet definition.** Packet size α counts spikes. Packet spread σ measures their timing dispersion. The curves illustrate why equal-sized packets need not have the same effect. They are not simulated data.

Generates Result 1.

2. **Compare packets with different starting states.** Use one fixed background realization and discard the first 30 ms as settling time. Tune the background so every pool

fires between 8 and 12 Hz per neuron. Test feedforward strengths 0.10, 0.12, and 0.14 μS . Select the weakest point at which an 80-spike, 3 ms reference packet reaches every pool while background-only coincidences remain below the packet threshold. Freeze those settings. Then test a weak diffuse packet, a broad strong packet, and a narrow oversized packet. Detect a packet when its 1 ms peak exceeds that pool's largest background-only peak by at least five spikes. Measure α as the number of neurons participating within 3 ms of that peak and calculate their temporal spread σ .

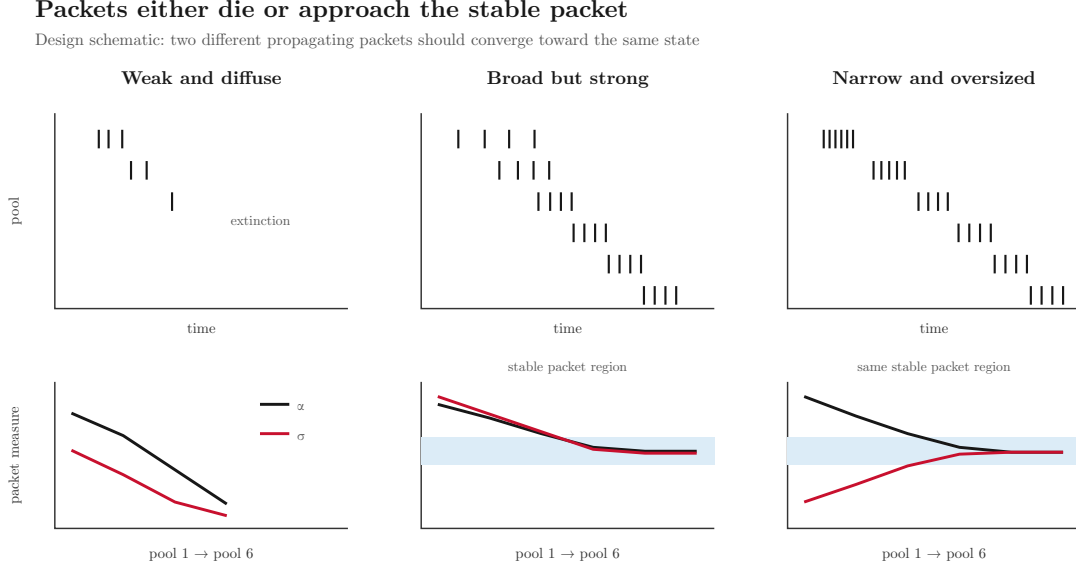


Figure 3: **Expected packet fates.** A weak or diffuse packet should die. Two different packets inside the propagation regime should approach the same stable size and spread. These trajectories explain the comparison and are not simulated data. Result 2 reproduces this layout with measured spikes and packet statistics.

Generates Result 2.

3. **Map the packet state transformation.** Keep the tuned network and background input fixed. Test 12 packet sizes from 20 to 100 spikes and seven temporal spreads from 0.2 to 5 ms. At each pool, plot the measured packet state (α_n, σ_n) and draw an arrow to the state produced in the next pool, $(\alpha_{n+1}, \sigma_{n+1})$. Classify a packet as extinct when no later pool produces a response volley. Locate the boundary between extinction and propagation, and test whether surviving trajectories approach one common state.

The chain should contain an extinction boundary and a packet attractor

Design schematic: arrows show the layer-to-layer transformation of packet state

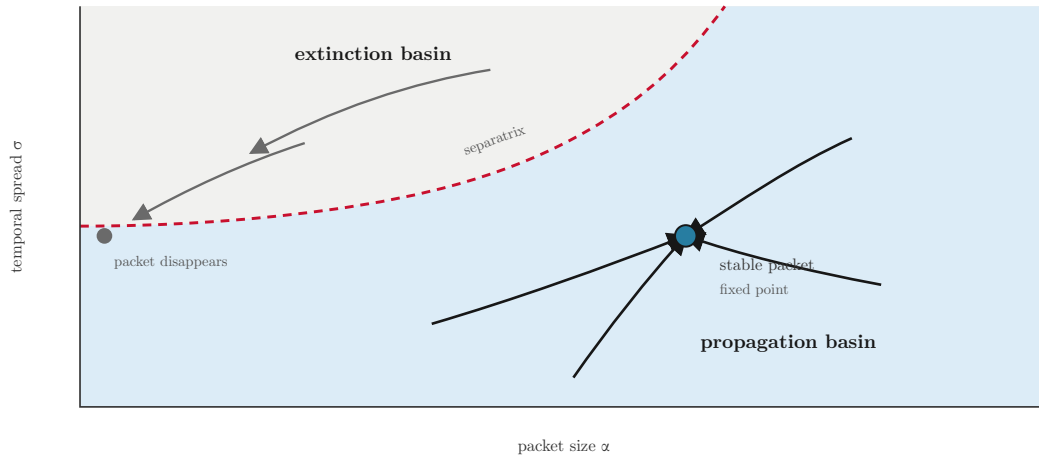


Figure 4: **Expected packet state space.** The dashed line separates initial packets that disappear from those that propagate. Layer-to-layer arrows in the propagation basin should approach one stable packet. This is a design schematic, not simulated data. Result 3 shows the measured transformations.

Generates Result 3.

4. **Render a looping raster hero.** Use the complete measured broad-packet run from Method 2. After the 30 ms settling period, scroll a 40 ms window through the remaining activity. Draw every measured spike as a conventional raster tick on a white field. Show background spikes in black, spikes inside the measured packet windows in cyan, and packet-input time as a red dashed line. Repeat the measured 30–100 ms segment to make an eight-second seamless loop. The repeated boundary is an animation device; the spikes within the segment come from one simulation.

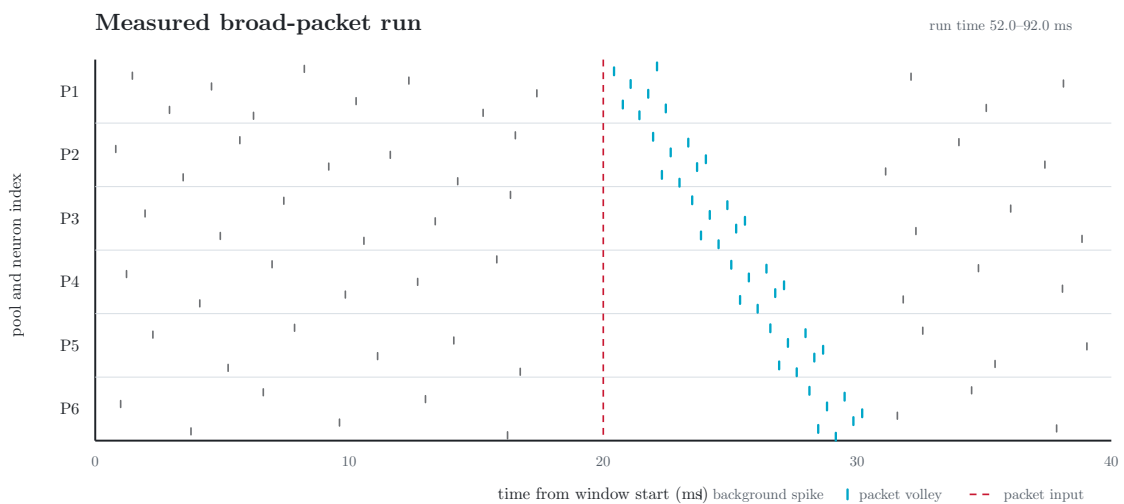


Figure 5: **Design frame for the hero animation.** Result 4 replaces these illustrative marks with every measured spike from the broad-packet run.

Generates Result 4.

Results

1. Reference packet propagation.

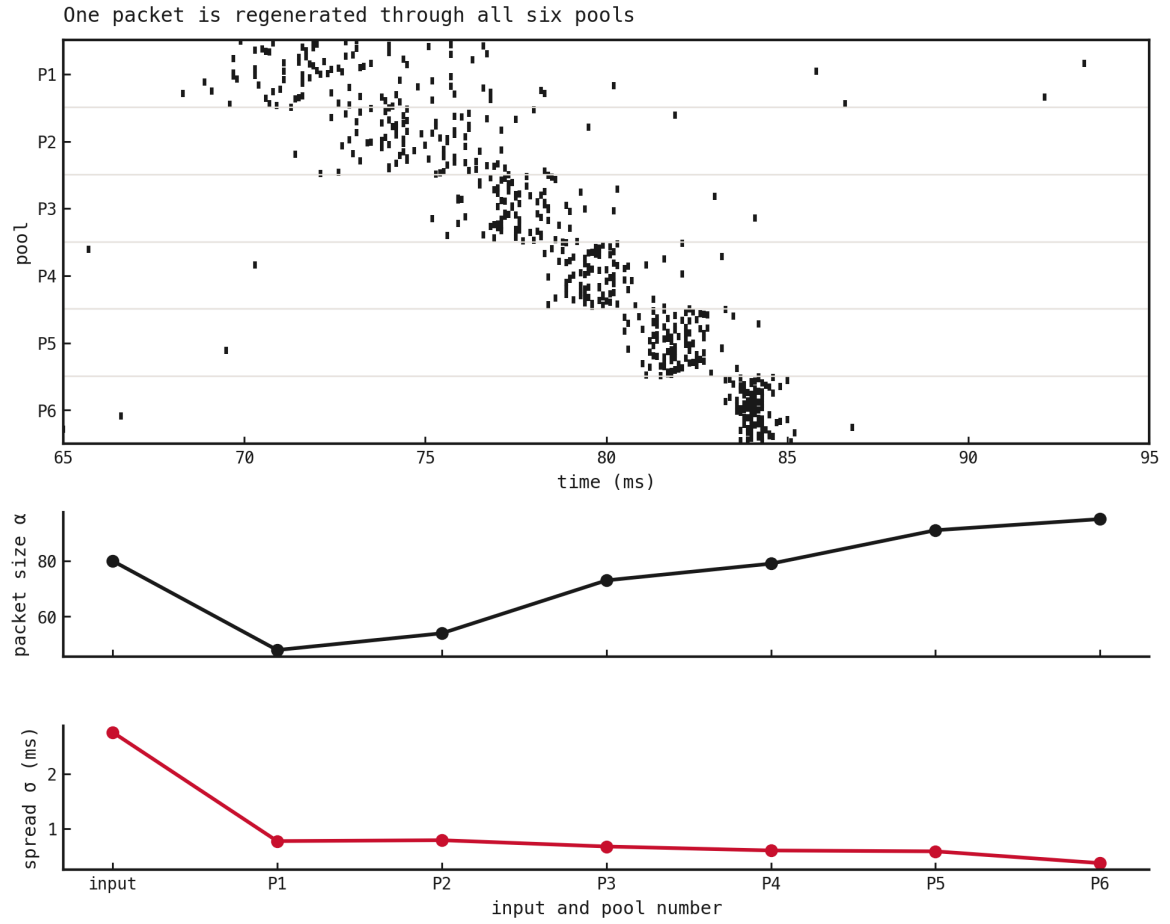


Figure 6: Reference-packet propagation at total feedforward strength $0.12 \mu\text{S}$. External background channels fire at 100 Hz, producing 10.07 Hz mean neural output.

The broad reference packet recruits 48 neurons in Pool 1 and 54 in Pool 2. It reaches 95 neurons in Pool 6 while narrowing from 0.78 ms to 0.37 ms. Each pool regenerates one coherent volley above its background activity.

2. Three packet trajectories.

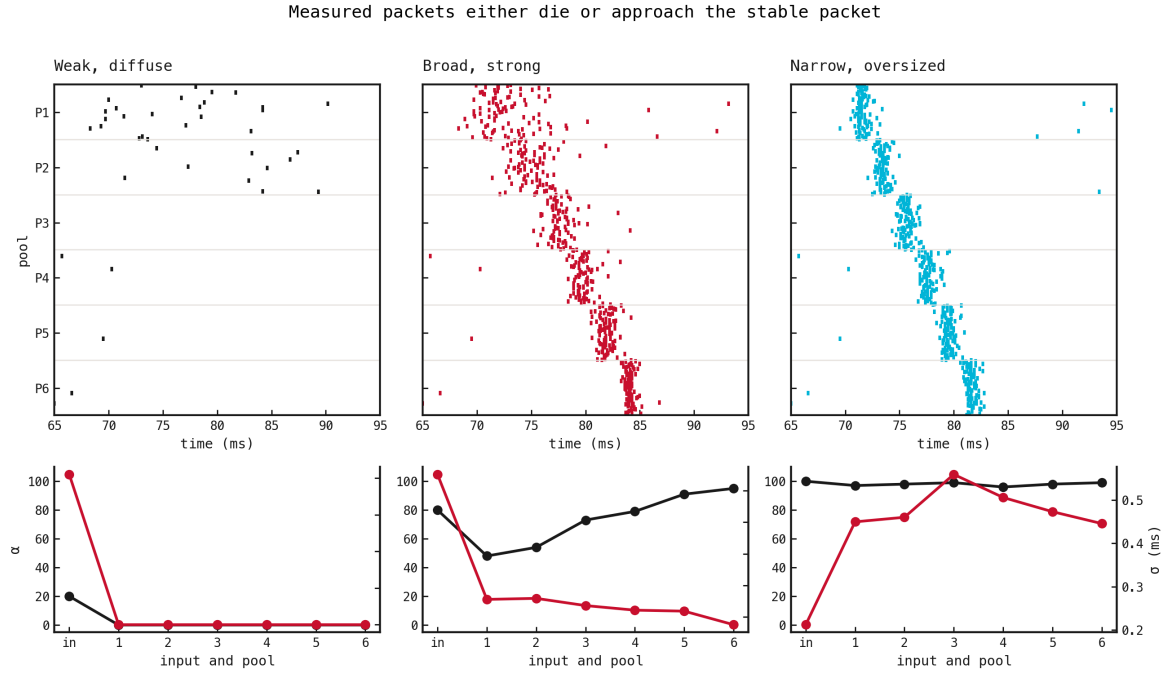


Figure 7: Measured packet fates. The upper panels show spikes grouped by pool. The lower panels show packet size α in black and spread σ in red.

The weak input produces no volley distinguishable from background. The broad and oversized packets reach 95 and 99 participating neurons, with late-chain spreads of 0.37 and 0.45 ms. Their final states are close despite very different initial temporal spreads.

3. Packet state space.

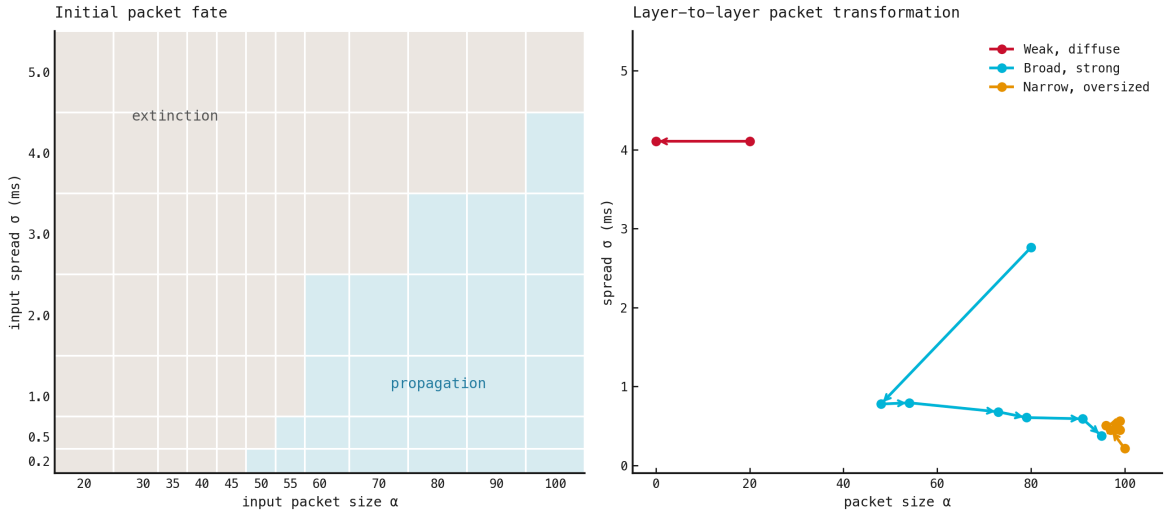


Figure 8: Measured packet state space. Tiles classify 84 initial size-width combinations. Arrows follow the three representative packets from their inputs through successive pools.

27 of 84 tested packets reach Pool 6. The boundary shifts toward larger α as σ increases: temporally diffuse packets require more spikes. Surviving trajectories approach a high-participation packet, while the weak trajectory moves directly to extinction.

4. Looping raster hero.

[Video — Measured spikes from the three packet trajectories, arranged as an eight-second seamless scrolling raster. · view the web edition to play.]

Black marks are genuine background-driven spikes from the measured run. Cyan marks are the same measured spikes when they fall inside a detected packet volley. The red dashed line marks input time. The raster repeats the settled 70 ms segment only to form a seamless loop; it does not splice different simulations together.

References

References

1. Diesmann, M., Gewaltig, M.-O., and Aertsen, A. (1999). Stable propagation of synchronous spiking in cortical neural networks. *Nature* 402, 529–533. doi:10.1038/990101